

INTRODUCTION

Three distinct datasets are analysed in this coursework. The Dutch boy's BMI dataset is analysed in the first section using the GAMLSS (Generalised Additive Models for Location, Scale, and Shape) distribution fit. The second section generates the centile curves for a given age after analysing the handgrip (HG) strength of English schoolchildren. The final dataset will be an analysis of a lab experiment's concrete compressive strength. The code loads the tidyverse, MASS, and gamlss packages first. It loads the DBBMI dataset, which includes information on body mass index (BMI) and other variables, from the gamlss.data package. The BMI data is extracted from this subset and put into a variable named bmi14.

Lastly, it uses the hist () function to generate a histogram of the bmi14 data, which shows the distribution of BMI values among people in the 14–15 age range.

The second dataset shows how English schoolchildren's handgrip (HG) strength relates to their age and gender. Cohen & associates (2010). Centile curves for the grip given age will be created by analysing 1000 distinct samples. The variables age and grip are contained in the data, which is saved in the galmss.data package with the name grip.

The Third dataset was obtained from the Kaggle to analyze the relationship between (mpg) which stands for miles per gallon and is used as the response or dependent variable and it was determined from laboratory experiments. This data has 32 obs. of 11 variables. When utilising the MT cars dataset for modelling, "mpg" is frequently utilised as the response or dependent variable. This variable is frequently the one attempting to forecast or explain using data from other predictor variables, like "wt." (weight), "hp" (horsepower), and so on.

FIRST DATA SET

[A]

```
> library(gamlss)
> library(tidyverse)
> data("dbbmi")
> old <- 14
> da <- with(dbbmi, subset(dbbmi, age > old & age < old + 1))
> bmi14 <- da$bmi
> hist(bmi14)
```

FIGURE: 1

This code loads a dataset with BMI data, picks, and extracts data for 14-year-old individuals, and builds a histogram to show the distribution of BMI for these individuals.

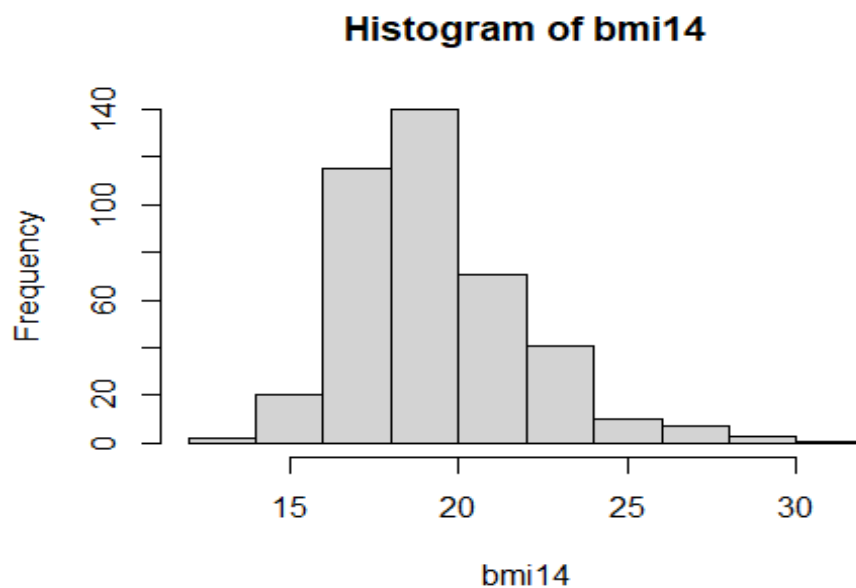


FIGURE:2

Data for Dutch boys aged 0 to 21 is included in the dataset, which is sourced from the BMI. This coursework will focus on the ages of 13 and 14. The age range was extracted by first setting the cut off age and then subsetting the dbbmi dataframe using the with() function. The plot of the histogram was made using the truehist() function.

```
> summary(dbbmi)
      age      bmi
Min.   : 0.030  Min.   :11.17
1st Qu.: 1.863  1st Qu.:15.96
Median :10.450  Median :17.45
Mean   : 9.291  Mean   :18.03
3rd Qu.:15.130  3rd Qu.:19.60
Max.   :21.700  Max.   :35.42
```

FIGURE: 3

These summary statistics aid in comprehending the central tendency, spread, and range of values for each variable by giving a succinct picture of the distribution of ages and BMIs in the "dbbmi" dataset.

[B]

NORMAL DISTRIBUTION: The probability distribution known as the normal distribution, or Gaussian distribution, is symmetric about the mean and indicates that data close to the mean occur more frequently than data far from the mean. A graph of the normal distribution shows up as a "bell curve".

A probability bell curve is correctly referred to as the normal distribution.

The mean and standard deviation of a normal distribution are both equal to 1. It has a kurtosis of three and zero skew.

While symmetrical distributions can be any distribution, not all symmetrical distributions are normal.

LOGNORMAL DISTRIBUTION: The distribution log normally displays a normal distribution curve's log of random variables. The exponent to which a base number must be raised to obtain the random variable (x) that is found along a normally distributed curve is generally referred to as the log.

GAMMA DISTRIBUTION: The gamma distribution is most used to simulate the interval between independent events that happen at a steady average rate. Analysts can model the number of events, such as the time until the second or third accident occurs, using this distribution. Reliability analysts model failure times in this situation using the gamma distribution.

EXPONENTIAL DISTRIBUTION: The exponential distribution is a continuous probability distribution that frequently relates to the remaining time before a particular event occurs. It's a process where things happen on their own, continuously, and at a steady average pace. Memorylessness is a crucial characteristic of the exponential distribution. Either more small values or fewer larger variables can make up the exponential random variable.

```
> # Fit different distributions
> fit_normal <- fitdistr(bmi14, "normal")
> fit_lognormal <- fitdistr(bmi14, "lognormal")
> fit_exponential <- fitdistr(bmi14, "exponential")
> fit_gamma <- fitdistr(bmi14, "gamma")
> # calculate AIC for each distribution
> aic_values <- c(AIC(fit_normal), AIC(fit_lognormal), AIC(fit_gamma), AIC(fit_exponential))
> names(aic_values) <- c("Normal", "Log-Normal", "Gamma", "exponential")
```

FIGURE: 4

I used the analysis's lowest-ranking classification (AIC) to describe the BMI data for fourteen-year-old Dutch boys. Model selection is done with the help of a statistical measure known as the AIC, whose lower values denote better fit. Gaining knowledge of each distribution's goodness-of-fit was obtained by comparing the AIC values of various classes. An in-depth understanding of each distribution's suitability for capturing the underlying patterns in the BMI dataset is provided by the AIC values associated with the normal classification. To facilitate informed analysis and interpretation, this thorough evaluation ensures that the chosen classification accurately captures the distributional properties of the data.

```
> # print AIC values
> print(aic_values)
```

Normal	Log-Normal	Gamma	exponential
1962.881	1921.697	1932.805	3251.086

FIGURE: 5

Since the Q-Q plot demonstrates a good fit and the log-normal distribution has the lowest AIC value, I determine that it is the most suitable model for the BMI data. The corresponding log-normal distribution map is shown. The normal log-normal distribution is a good fit for explaining the distribution of BMI in this demographic subset because it effectively captures the central tendency and variability of BMI values.

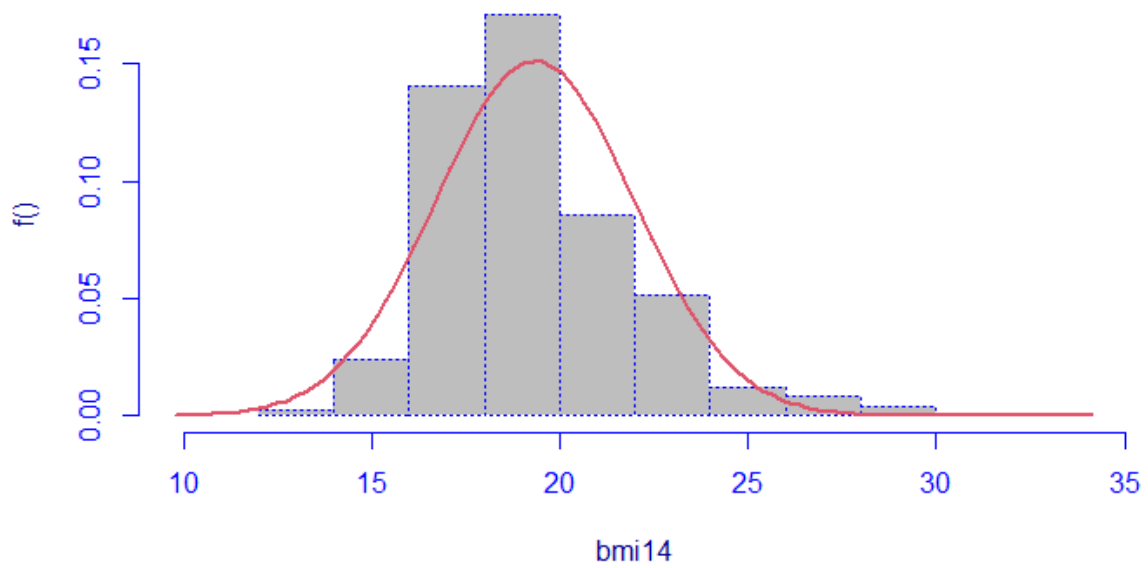


FIGURE: 6

[C]

Various parametric distributions are fitted to the data, and a suitable distribution is selected by:

- Identifying the Distributions of Candidates: Determine candidate distributions that are frequently employed to model BMI data first. Several typical distributions for continuous. The normal (Gaussian), log-normal, gamma, and Exponential distributions are among the data types.
- Distributions Fit to Data: I make the distributions fit to the BMI data. I have access to statistical packages that offer fitting methods such as maximum likelihood estimation.
- Assess Fit: Following the fitting of each distribution, I assess the fit using suitable statistical metrics, such as the Bayesian Information Criterion (BIC), the Akaike Information Criterion (AIC), or visual examination of quantile-quantile (Q-Q) plots.

```
-----
No. of observations in the fit: 410
Degrees of Freedom for the fit: 2
    Residual Deg. of Freedom: 408
        at cycle: 2

Global Deviance:    1917.697
      AIC:          1921.697
      SBC:          1929.73
*****
```

FIGURE: 7

```

> fitlognorm <- fitdistr(bmi14, "lognormal")
> print(fitlognorm)
      meanlog      sdlog
 2.953531499  0.130835465
(0.006461503) (0.004568973)

```

FIGURE: 8

The summary result of the final chosen Log-normal model. The model yielded the estimated parameter values as follows: the average is 2.9535, and the SD log is 0.1308.

The parameters for the mean and standard deviation of the log-normal distribution are these

values, correspondingly. The mean log parameter, which is the logarithm of the median BMI value, represents the spatial parameter of the distribution, while the SD log parameter represents the scale parameter, or the spread or variation of the BMI values along the median.

Key parameter values that are essential for comprehending the properties of the distribution are revealed in the summary output of the finalised Log-normal model. The Mean log parameter, which represents the spatial component of the distribution, is found to be 2.95353 based on the model estimation.

SECOND DATA SET

[A]

In order to analyse the secondary data, three different models are used: the Box-Cox Cole and Green (BCCG), Box-Cox t (BCT), and Box-Cox Power Exponential (BCPE). There are advantages and considerations unique to each model. The grip strength data's skewed and heavy-tailed nature makes it especially suitable for the BCCG model, which uses the Cole and Green power index distributions. The BCT model generates accurate centile estimates and is robust against heteroscedasticity and outliers. To achieve this, it transforms the response variables using the Box-Cox method and the errors using the t distribution method. Lastly, the BCPE model incorporates a Box-Cox adjustment on the scale parameter and response, allowing for flexible modelling of the age-grip strength connection under various assumptions.

[B]

Following the instructions, we selected a subset of 1000 English boys using a distinct seed number (320) from the 3766-item original dataset. This is helpful for reproducibility because it guarantees that the random sampling carried out in the next step yields the same result each time the code is run.

Using the sample () function, random sampling It takes a random sample of 1000 integers, ranging from 1 to 3766. These numbers are probably row numbers or indices. The variable index contains this random sample.

Subset Formation: It uses the random indices created in the previous step to subset a dataset called grip. To be more precise, it takes the rows that match the randomly chosen indices (index) and saves this subset in a brand-new dataset called my data.

```
> set.seed(320)
> index<-sample(3766,1000)
> mydata<-grip[index,]
> dim(mydata)
[1] 1000    2
```

FIGURE: 9

The handgrip strength (HG) of English schoolchildren was examined in this dataset in relation to age and gender. There are 3766 data points in the dataset. The information is available via the grip name in the galmss.data library. Age and grip are the variables in the dataset. The dataset's primary goal is to produce centile curves for grip for each age. Using the sample () function, 1000 samples will be chosen for this coursework.

[C]

All in all, this code creates a line plot from my dataset, with the grip variable represented on the y-axis and the age variable on the x-axis. A line that connects the data points at each point in the plot shows the trend or relationship between grip strength and age.

```
> plot(mydata, xlab="age", ylab="grip")
```

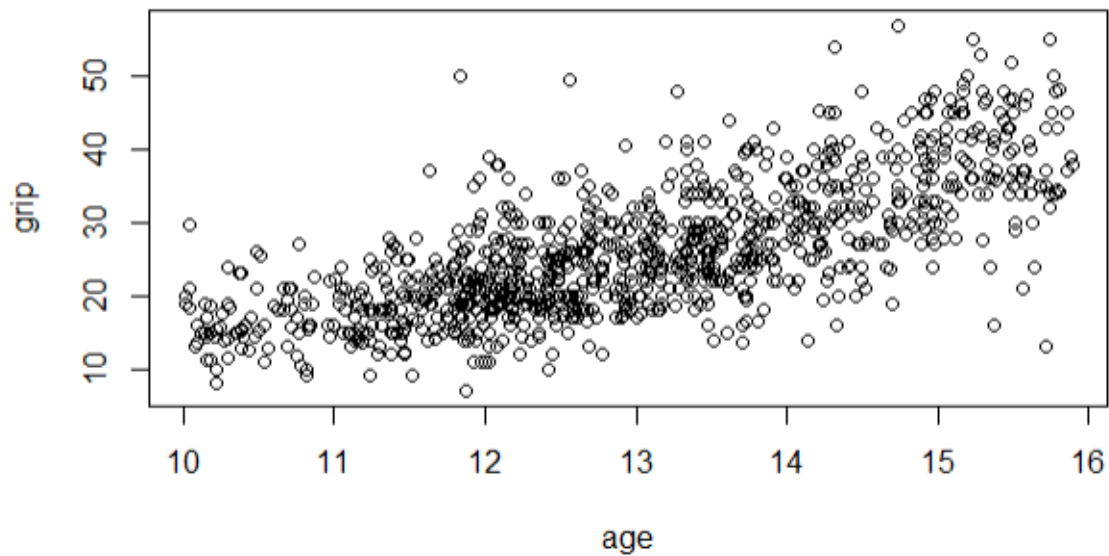


FIGURE: 10

The scatter plot above displays the relationship between grip strength and age. Given that power transform is used for nonlinear relationships in data, there is no need to power transform the age in this set because age and grip have a linear relationship.

The LMS (lamda, mu, and sigma) was developed by Cole and Green to create the centile curve for the response variable against the explanatory variable. The second data, which the LMS approach fits, now fits the Box-Cox-Cole and Green distribution (BCCG).

It's important to think about whether the age variable needs to be transformed when plotting grip against age. Age can often be plotted straight-forwardly without the need for any power transformation. There are various reasons:

- Age is a characteristic that typically advances in a linear fashion over time. For example, grip strength may vary gradually with age, with no abrupt changes or nonlinear patterns. Thus, this linear relationship can be adequately represented by plotting grip against age directly.
- The plot remains interpretable if the age variable is kept on the original scale. Anyone watching the plot will find it more difficult to interpret the x-axis when age is transformed.
- The interpretation of the relationship between variables becomes more complex when power transformations are used. You keep the analysis more straightforward and comprehensible by not changing the age.

Prior to drawing any conclusions, it is usually a good idea to visually analyse the relationship between the variables. One can thus directly observe any linear or nonlinear patterns in the data by plotting grip against age without power transformation, and then make defensible decisions based on that observation.

```

Family: c("NO", "Normal")
Fitting method: "nlminb"

Call: gamlssML(formula = bmi14, family = "NO")

Mu Coefficients:
[1] 19.34
Sigma Coefficients:
[1] 0.9699

Degrees of Freedom for the fit: 2 Residual Deg. of Freedom 408
Global Deviance: 1958.88
AIC: 1962.88
SBC: 1970.91

```

FIGURE: 11

To compare how well the BCCG, BCT, and BCPE distributions fit the data, the value obtained from fitting the model presented was utilised.

An explanation of the function and the result is provided below:

Mu Coefficients: Gives the approximate coefficient for the distribution's mean, or location parameter. The estimated mean in this instance is 19.34.

Sigma Coefficients: These give the approximate coefficient for the distribution's scale parameter, or standard deviation. It is roughly 0.9699 here.

Graduations of Deviance and Freedom:

Degrees of Freedom: Denotes both the residual and fit-related degrees of freedom.

```

> gbccg <-gamlss(grip ~pb(age),sigma.fo = ~pb(age),nu.fo=~pb(age),data = mydata,family = BCCG)
GAMLSS-RS iteration 1: Global Deviance = 6365.646
GAMLSS-RS iteration 2: Global Deviance = 6352.245
GAMLSS-RS iteration 3: Global Deviance = 6351.753
GAMLSS-RS iteration 4: Global Deviance = 6351.727
GAMLSS-RS iteration 5: Global Deviance = 6351.725
GAMLSS-RS iteration 6: Global Deviance = 6351.725

```

FIGURE: 12

The `edfAll()` function shall be used to obtain the degree of freedom for each parameter below.


```

> edf(gbccg)
Effective df for mu model
$`pb(age)`
[1] 4.525578

> edfAll(gbccg)
$mu
$mu$`pb(age)`
[1] 4.525578

$sigma
$sigma$`pb(age)`
[1] 2.002879

$nu
$nu$`pb(age)`
[1] 2.660247

```

FIGURE: 13

From the BCCG distribution degree of freedom for mu, sigma and nu are 4.52, 2.00 and 2.66 respectively.

Since the starting point is the fitted values from the LMS model(BCCG model) by the four parameter distributions BCT and BCPE as the BCCG lacks the tau parameter as it is a three parameter distribution. The BCT and BCPE was fitted using p-spline smoother.

[E]

```

> gbct <-gamlss(grip~pb(age),sigma.fo=~pb(age),nu.fo=~pb(age),tau.fo=~pb(age),data = mydata,family
= BCT,start.from = gbccg)
GAMLSS-RS iteration 1: Global Deviance = 6340.792
GAMLSS-RS iteration 2: Global Deviance = 6339.112
GAMLSS-RS iteration 3: Global Deviance = 6338.686
GAMLSS-RS iteration 4: Global Deviance = 6338.529
GAMLSS-RS iteration 5: Global Deviance = 6338.469
GAMLSS-RS iteration 6: Global Deviance = 6338.446
GAMLSS-RS iteration 7: Global Deviance = 6338.436
GAMLSS-RS iteration 8: Global Deviance = 6338.433
GAMLSS-RS iteration 9: Global Deviance = 6338.432
GAMLSS-RS iteration 10: Global Deviance = 6338.432

```

FIGURE: 14

```

> edf(gbct)
Effective df for mu model
$`pb(age)`
[1] 4.61928

> edfAll(gbct)
$mu
$mu$`pb(age)`
[1] 4.61928

$sigma
$sigma$`pb(age)`
[1] 2.562621

$nu
$nu$`pb(age)`
[1] 3.109525

$tau
$tau$`pb(age)`
[1] 2.000027

```

FIGURE: 15 (Degrees of freedom for BCT)

The LMS model's fitted values (gbccg) served as a basis for fitting the dataset to the BCPE and BCT distributions. The models' ability to adjust to changes in grip strength brought on by ageing is demonstrated by their effective degree of freedom. Greater adaptability to the intricacies

A higher value indicates data. The models' respective effective degree of freedom values indicated that there was comparable variability in how well they captured the relationship between alterations brought on by ageing and grip strength. A visual representation of these outcomes is provided.

```

>
>      gbcp <- gamlss(grip ~ pb(age),
+                    sigma.fo = ~pb(age),
+                    nu.fo = ~pb(age),
+                    lambda.fo = ~pb(age),
+                    data = mydata,
+                    family = BCPE,
+                    start.from = gbccg)
GAMLSS-RS iteration 1: Global Deviance = 6343.511
GAMLSS-RS iteration 2: Global Deviance = 6343.063
GAMLSS-RS iteration 3: Global Deviance = 6343.064
GAMLSS-RS iteration 4: Global Deviance = 6343.086
GAMLSS-RS iteration 5: Global Deviance = 6343.093
GAMLSS-RS iteration 6: Global Deviance = 6343.096
GAMLSS-RS iteration 7: Global Deviance = 6343.096
>
> edf(gbcp)
Effective df for mu model
$`pb(age)`
[1] 4.555739

```

FIGURE: 16

[F]

The models that were compared using the generalised Akaike information criterion (GAIC) were BCCG, BCT, and BCPE. After utilising the dataset to calculate each model's GAIC values, the following conclusions were drawn.

```

> cat("GAIC for BCCG model:", gaic_bccg, "\n")
GAIC for BCCG model: 6370.102
> cat("GAIC for BCT model:", gaic_bct, "\n")
GAIC for BCT model: 6363.015
> cat("GAIC for BCPE model:", gaic_bcpe, "\n")
GAIC for BCPE model: 6364.412

```

FIGURE: 17

Following analysis, the optimal option was determined to be the BCT model with the lowest GAIC value of 6363. This demonstrates that the BCT model offers the best balance between model robustness and goodness of fit among the models examined. The BCCG, BCT, and BCPE models' GAIC values are presented graphically, which also emphasises the model selection procedure based on comparison.

```

> if (gaic_bccg < gaic_bct & gaic_bccg < gaic_bcpe) {
+   cat("BCCG model has the lowest GAIC and is the preferred model.\n")
+ } else if (gaic_bct < gaic_bccg & gaic_bct < gaic_bcpe) {
+   cat("BCT model has the lowest GAIC and is the preferred model.\n")
+ } else {
+   cat("BCPE model has the lowest GAIC and is the preferred model.\n")
+ }
BCT model has the lowest GAIC and is the preferred model.

```

FIGURE: 18

The fitted parameters for BCT, BCPE and BCCG can be plotted by using the fitted plot() function.

```

> fittedPlot(gbccg,gbct, gbcp,x=mydata$age)

```

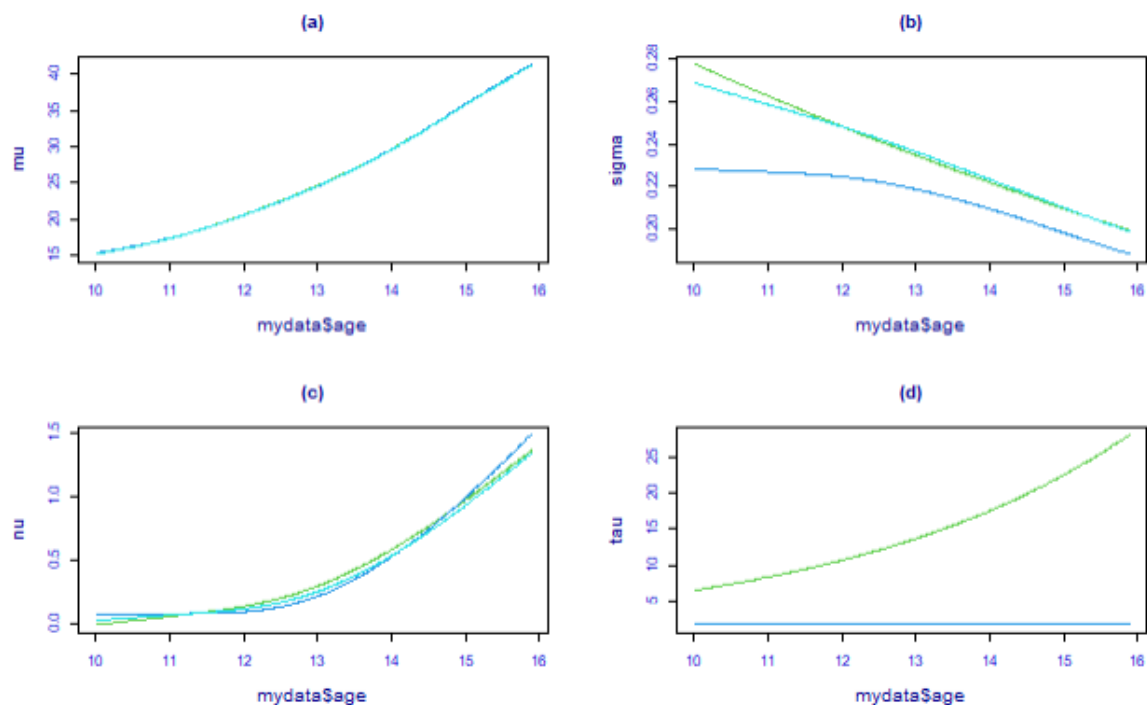


FIGURE: 19

[G]

```

> fittedPlot(gbccg,gbct,x=mydata$age)

```

Comparison of the BCT and BCCG models using GAIC

The model BCT has the lowest AIC value which means it is the best model when compared to the BCCG. A lower GAIC value indicates a better trade-off between model number of parameters and goodness of fit. The fitted parameters for BCT and BCCG can be plotted by using the `fittedplot()` function.

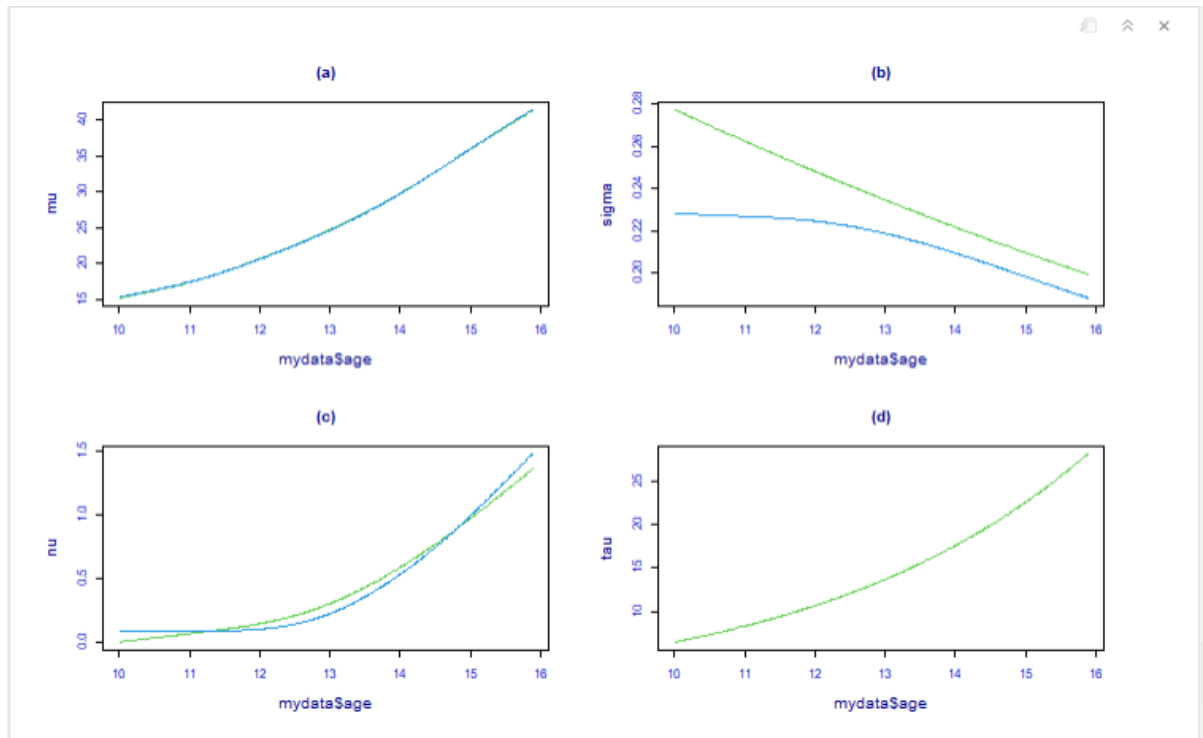


FIGURE: 20 (Fitted value for μ , σ , τ , and ν parameters against age for BCT and BCCG models.)

[H]

Using the `centiles()` function, centile analysis of the fitted models (BCCG and BCT) for grip strength evaluation was performed, yielding the fraction of instances falling below specific centile thresholds. These calculated centile values were then stored in variables so they could

be put into comparison. We can then evaluate the accuracy of each model's grip strength prediction at various ages by examining these centile values. To visualise this comparison analysis and assess how well the BCCG and BCT models performed in determining grip strength at different centile thresholds within the population, a centile plot was employed. Ultimately, this study enabled selection of a model based on its ability to accurately depict the distribution of grip strength.

```

> centiles_gbccg <- centiles(gbccg)
% of cases below 0.4 centile is 0.7
% of cases below 2 centile is 2.7
% of cases below 10 centile is 8.5
% of cases below 25 centile is 23.4
% of cases below 50 centile is 51.9
% of cases below 75 centile is 75.8
% of cases below 90 centile is 90.5
% of cases below 98 centile is 97.9
% of cases below 99.6 centile is 99.3

```

FIGURE: 21

```

> centiles_gbct <- centiles(gbct)
% of cases below 0.4 centile is 0.4
% of cases below 2 centile is 2.7
% of cases below 10 centile is 9.6
% of cases below 25 centile is 24.6
% of cases below 50 centile is 51.8
% of cases below 75 centile is 74.8
% of cases below 90 centile is 89.9
% of cases below 98 centile is 98.3
% of cases below 99.6 centile is 99.5

```

FIGURE: 22

```

> centiles_gbcg <- centiles(gbcg)
% of cases below 0.4 centile is 0.5
% of cases below 2 centile is 2.6
% of cases below 10 centile is 9
% of cases below 25 centile is 24.5
% of cases below 50 centile is 51.7
% of cases below 75 centile is 74.4
% of cases below 90 centile is 90.1
% of cases below 98 centile is 98.3
% of cases below 99.6 centile is 99.5

```

FIGURE: 23

The corresponding colours for the models—BCCG, BCT, BCPE, and fitted distributions—are light blue, green, and dark blue. All the models' mu coefficients are plotted against age in the first graph, denoted by "a." Each line overlaps the other, indicating that the medians of all the models are comparable.

centile curves were plotted using the centiles function. The centile is displayed in the lines at a given percentage. Every model has a comparable mean (i.e., 50% centile). Though it varies slightly for each, the 10th centile curve is nearly the same for all three distributions. The curve's shape is nearly uniform across all distributions at the 90th centile, and the centile is skewed slightly to the right. In relation to the 99.6th centile, the BCT model exhibits a marginally elevated centile curvature in comparison to the BCCG and BCPE models. These charts, which display the distribution of a specific growth measure (such as weight or height) at various ages, are used to monitor children's development. Every curve on the graph

denotes a distinct percentile, or the percentage of children in the population that fall below that value.

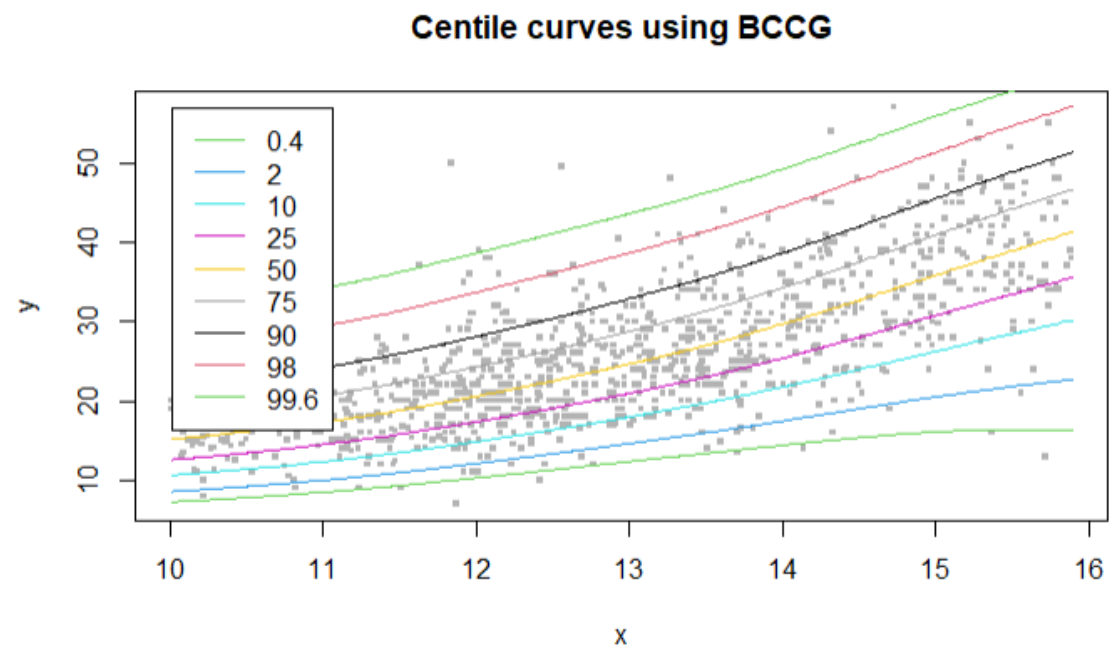


FIGURE: 24

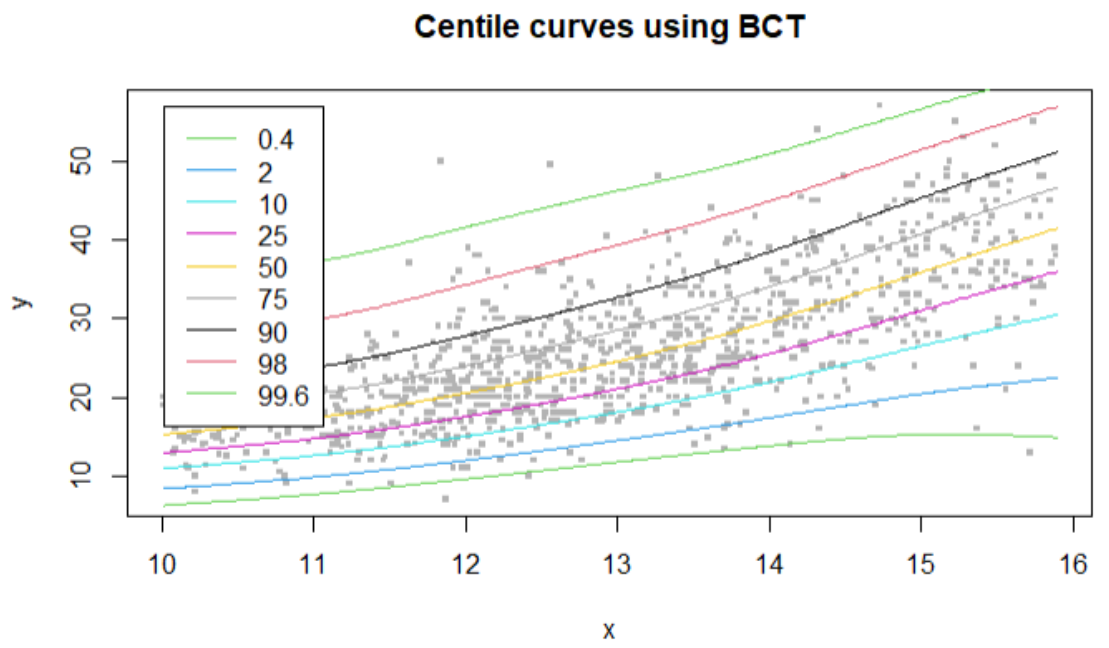


FIGURE: 25

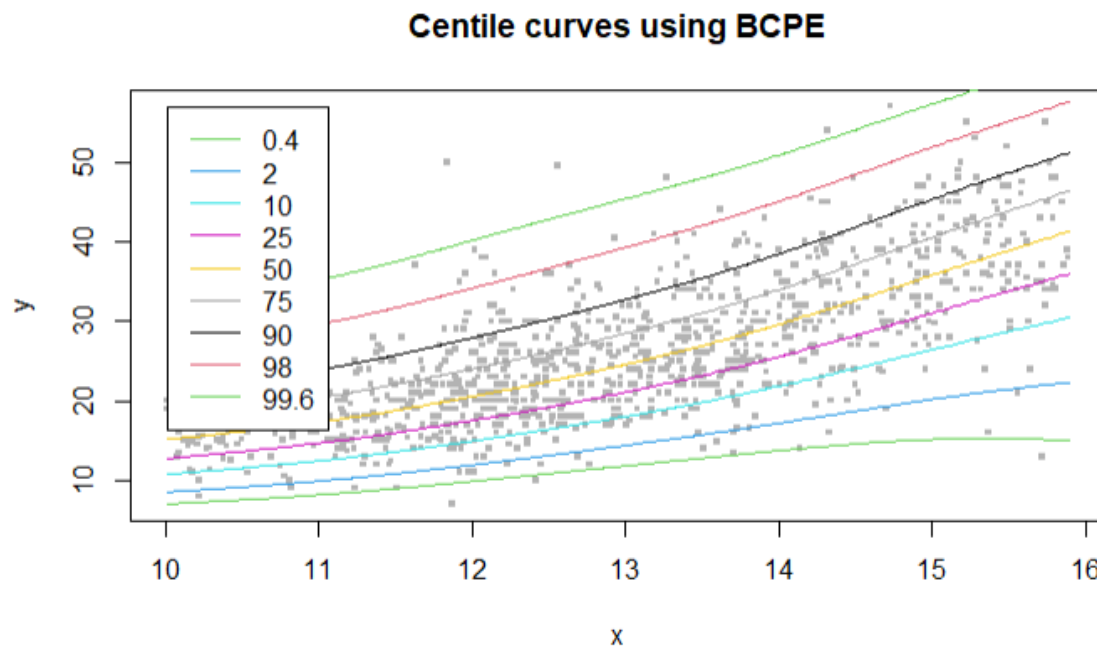


FIGURE: 26

[1]

```
> plot(gbccg, main = "BCCG Model Residuals")
*****
      Summary of the Quantile Residuals
              mean   = 0.001559423
              variance = 1.001039
      coef. of skewness = 0.009457053
      coef. of kurtosis = 3.567509
Filliben correlation coefficient = 0.9978855
*****
```

FIGURE: 27

plot of a BCCG model's residuals and supplied some summary data for those residuals. The following provides an explanation for every statistic:

Mean: The residuals have a mean of roughly 0.0016. In a well-fitted model, the mean of the residuals should ideally be near zero, meaning that the model is generally accurate in predicting the result.

Variance: The residuals have a variance of roughly 1.001. This quantifies the residuals' dispersion or spread around the mean. When the variance is near to 1, it indicates that the residuals are distributed uniformly around the mean.

Skewness coefficient: Skewness quantifies the asymmetry in the residual distribution. The distribution of residuals is roughly symmetrical if the coefficient of skewness is near zero.

The residuals' distribution's tailenders is measured by the coefficient of kurtosis. A residual distribution that is roughly normal is indicated by a coefficient of kurtosis that is near to 3, which represents the kurtosis of a normal distribution. Greater values signify heavier tails.

The linear correlation between the expected quantiles of a normal distribution and the observed quantiles of the residuals is measured by the Filliben Correlation Coefficient. The residuals are said to be normally distributed if their value is near to 1.

All things considered, these statistics are employed to evaluate a regression model's assumptions, especially the assumption of normally distributed residuals with constant variance. The more closely these numbers resemble the ideal circumstances (mean close to 0, variance close to 1, skewness and kurtosis close to 0 and 3, respectively), and Correlation coefficient of Filliben.

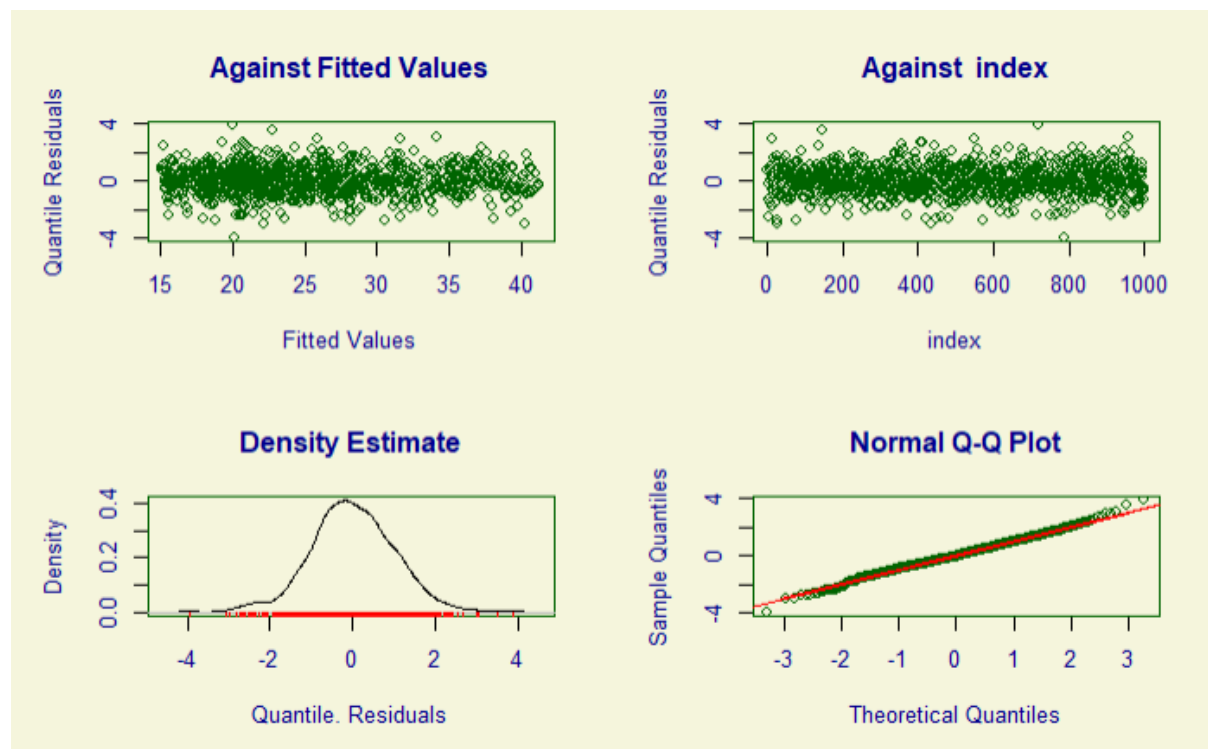


FIGURE: 28

```
> plot(gbct, main = "BCT Model Residuals")
*****
      summary of the Quantile Residuals
              mean   = 0.001241125
              variance = 1.001244
      coef. of skewness = -0.007111548
      coef. of kurtosis = 2.973848
Filliben correlation coefficient = 0.9993975
*****
```

FIGURE: 29

Approximately 0.00124 is the mean of the residuals. This model displays, like the previous one, the average difference between the observed and expected values. When the mean is near zero, it indicates that the model is generally doing a good job of forecasting the result.

Roughly 1.001 is the variance of the residuals. This measures the dispersion, or spread, of the residuals around the mean, just like it did previously. The residuals appear to be uniformly distributed around the mean when the variance is near 1.

A slight left skew is suggested by the skewness of about -0.007, which shows that the residual distribution is slightly skewed to the left. In a perfectly symmetric distribution, this suggests that there may be a small tendency for the residuals to be lower than the mean more frequently than would be predicted.

A small departure from the optimal kurtosis of three for a normal distribution is indicated by the kurtosis of roughly 2.97. This implies that the distribution's tails may be a little bit lighter or shorter than those of a normal distribution.

All things considered; these statistics indicate that the residuals derived from the BCT model generally conform to the linear regression assumptions quite well. Interpreting these statistics, however, in conjunction with other diagnostics and considering the particular context and analysis goals is crucial.

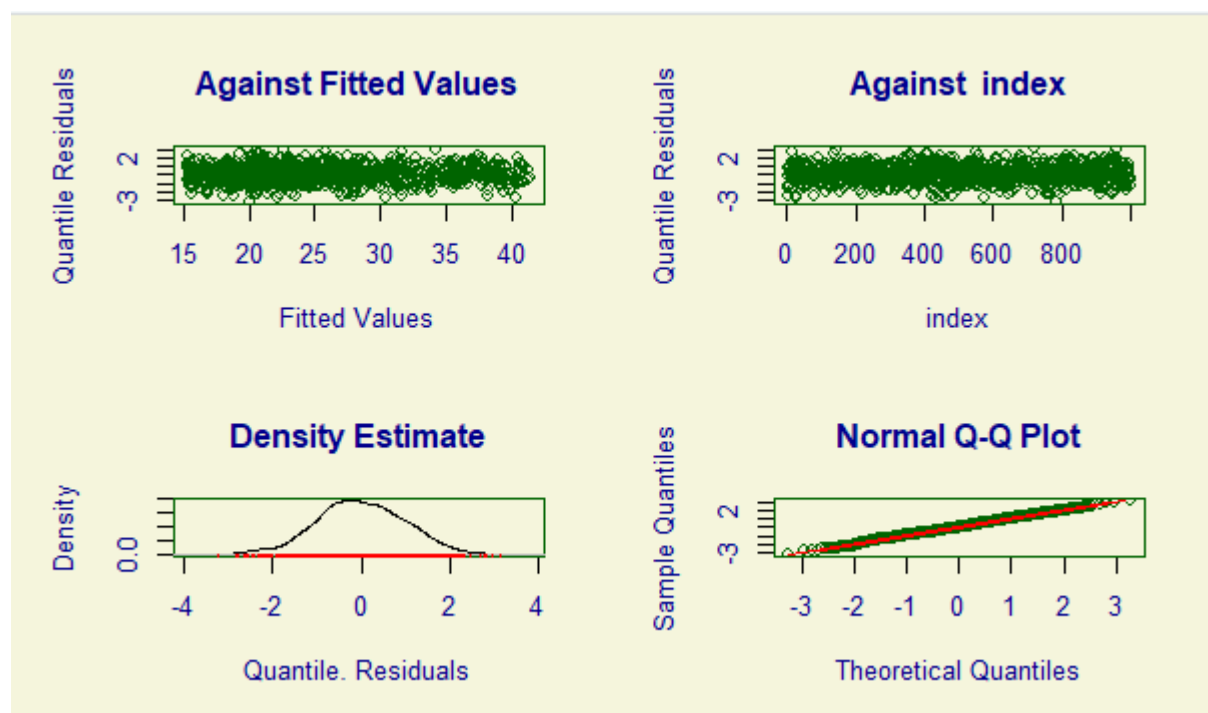


FIGURE: 30

```

> plot(gbcp, main = "BCPE Model Residuals")
*****
                Summary of the Quantile Residuals
                mean      =  0.006692742
                variance   =  1.000329
                coef. of skewness = -0.007196611
                coef. of kurtosis =  3.099792
Filliben correlation coefficient =  0.9991866
*****

```

FIGURE: 31

Mean: Approximately 0.00669 is the mean of the residuals. The average difference between the observed and expected values is shown here. A mean that is nearly equal to zero indicates that the model is generally forecasting the result with reasonable accuracy.

Variance: There is about a 1.000 variance in the residuals. This quantifies how widely distributed the residuals are around the mean. The residuals appear to be uniformly distributed around the mean when the variance is near 1.

Skewness Coefficient: The skewness coefficient of roughly -0.007 indicates a slight left skew, meaning that the residuals' distribution is slightly skewed to the left. This indicates that, as previously mentioned, there may be a small tendency for the residuals to be lower than the mean more frequently than anticipated.

Coefficient of Kurtosis: With a kurtosis of roughly 3.10, a normal distribution deviates slightly from the optimal kurtosis of 3. This implies that the distribution's tails could be a little bit longer or heavier than those of a normal distribution.

The degree of similarity between the expected quantiles of a normal distribution and the observed quantiles of the residuals is measured by the Filliben correlation coefficient.

Assuming that the residuals are roughly normally distributed, a value near 1 (0.9992 in this instance) denotes a strong similarity.

These figures imply that the BCPE model's residuals typically follow linear regression's presumptions fairly well. It is crucial to interpret these statistics in conjunction with other diagnostics and consider the particular context and objectives of the analysis, as is the case with all statistics.

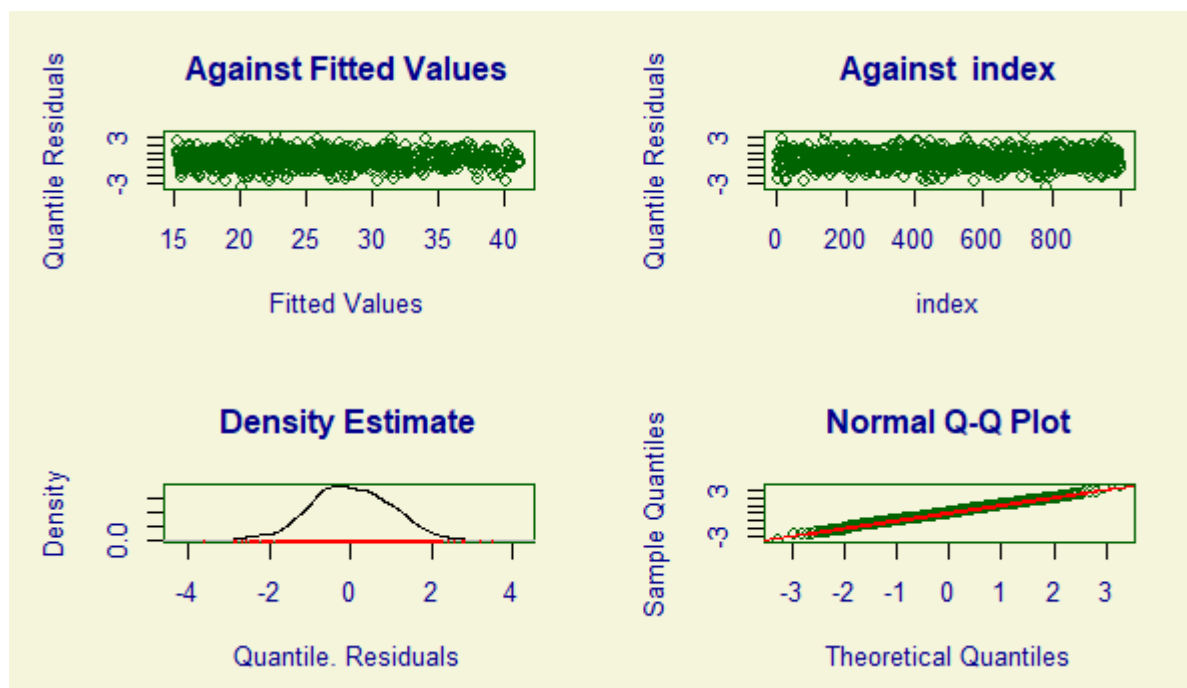


FIGURE: 32

WORM PLOTS

The distribution of data can be seen visually with a worm plot, among other plot types. The relationship between the data points and the quantiles—values that split the data into equal parts—is displayed. The quantiles displayed in the worm plots in this instance are Unit normal quantiles, which are quantiles derived from the conventional normal distribution. information about the three distinct distributions, BCCG, BCPE, and BCT. The "Deviation" section of each table displays three values for deviation (-0.4, 0.0, and 0.4), and the "Worm plot for [distribution name]" section shows a range of values along with a corresponding "Unit normal quantile."

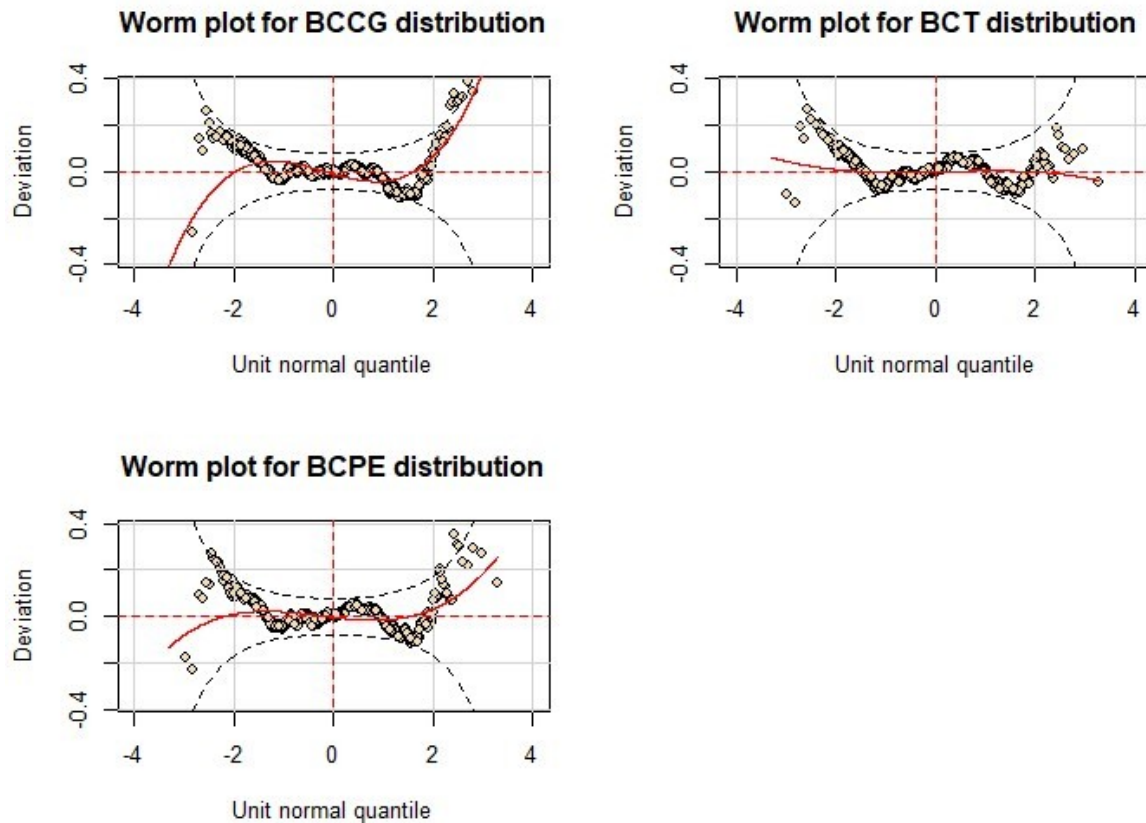


FIGURE: 33

Q – STATS

The function Q-Stats () provides location, scale parameter fits, skewness, and kurtosis. The residual Z Statistics plots for the BCCG, BCT, and BCPE models are shown.

The Z-statistic displays changing circle diameters and squares inside the circles.

The red and blue circles, each displaying a unique Z statistic value, represent Z1 through Z4 statistics. Larger circles correspond to higher absolute Z-values. The Z-statistics' magnitude is represented by the circles' sizes. A larger circle indicates a greater departure from normalcy over the same time.

Triangles that represent which way is up or down are located inside the circles. When the measured residuals in that interval are higher than what would be expected under normal conditions, the Z-statistic value is positive and is represented by an upward-pointing triangle. When a triangle represents a negative Z-statistic, it indicates that the observed residuals are lower than expected. These visual signals facilitate the identification and comprehension of periodic departures from normalcy by researchers and facilitate the comprehensive analysis of sample residues.

```
> q.stats(gbccg)
```

	Z1	Z2	Z3	Z4	AgostinoK2	N
0.5 to 100.5	-0.006159195	0.79941791	-1.55991869	1.45173781	4.540888991	100
100.5 to 200.5	0.584029866	-1.13451851	2.06598576	2.14725895	8.879018166	100
200.5 to 300.5	-1.605139816	-1.49463542	-0.19973220	-0.39725919	0.197707815	100
300.5 to 400.5	0.235029302	-0.59702055	1.46355522	0.08619531	2.149423520	100
400.5 to 500.5	-0.376911809	2.18935633	-0.53025633	0.45658861	0.489644926	100
500.5 to 600.5	0.290322282	-1.31733891	0.86063687	1.51491973	3.035677612	100
600.5 to 700.5	-0.141289648	-0.64534488	0.55885112	0.17974746	0.344623726	100
700.5 to 800.5	-0.581585633	1.47212995	-1.28132318	2.92616340	10.204221346	100
800.5 to 900.5	1.558769724	-0.79017076	1.31887977	-1.45691367	3.862041288	100
900.5 to 1000.5	0.198877241	0.85345577	0.07670789	0.01279090	0.006047708	100
TOTAL Q stats	6.026710781	14.98162590	13.60482952	20.10446558	33.709295098	1000
df for Q stats	5.474421706	8.49856045	7.33975280	10.00000000	17.339752795	0
p-val for Q stats	0.358316315	0.07428504	0.06908019	0.02827996	0.010524326	0

FIGURE: 34

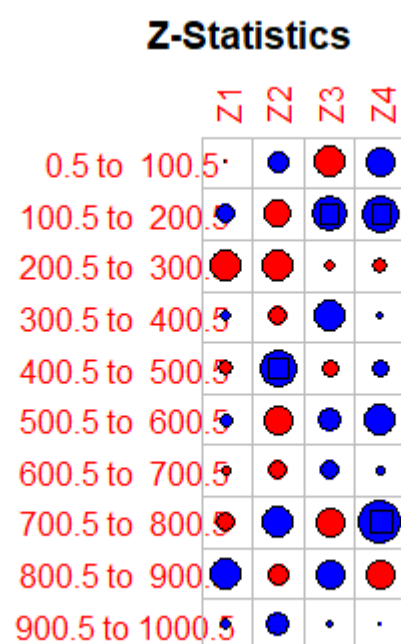


FIGURE: 35

```
> Q.stats(gbct)
```

		Z1	Z2	Z3	Z4	AgostinoK2	N
0.5 to	100.5	0.06557419	0.6904872	-1.35716779	0.6317983	2.2410735	100
100.5 to	200.5	0.51777135	-1.1411592	1.13224395	0.8104166	1.9387514	100
200.5 to	300.5	-1.62157106	-1.1986008	-0.05170806	-0.9557016	0.9160393	100
300.5 to	400.5	0.19393383	-0.4113326	1.26902202	-0.7277404	2.1400230	100
400.5 to	500.5	-0.39744515	2.0459933	-0.47453005	-0.3974874	0.3831750	100
500.5 to	600.5	0.25733067	-1.1870937	0.47385163	0.4967757	0.4713215	100
600.5 to	700.5	-0.13830111	-0.4269544	0.51348874	-0.3897089	0.4155437	100
700.5 to	800.5	-0.49528450	0.8687409	-1.20780696	1.6660529	4.2345299	100
800.5 to	900.5	1.52341617	-0.6281556	0.98719904	-2.3095947	6.3087895	100
900.5 to	1000.5	0.21868810	0.9354632	-0.04997683	-0.7459061	0.5588735	100
TOTAL Q stats		5.79672715	11.1868071	7.88621349	11.7219068	19.6081202	1000
df for Q stats		5.38072020	8.2186897	6.89047539	7.9999733	14.8904487	0
p-val for Q stats		0.37176941	0.2061688	0.33179678	0.1640481	0.1822860	0

FIGURE: 36

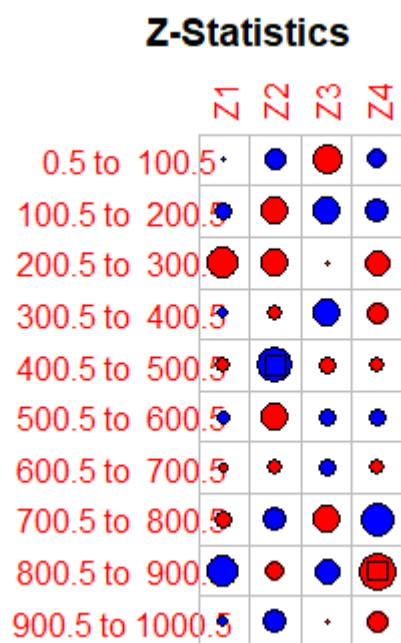


FIGURE: 37

```
> Q.stats(gbcp)
```

		Z1	Z2	Z3	Z4	AgostinoK2	N
0.5 to	100.5	0.1093658	0.6968285	-1.48257642	0.74555630	2.75388702	100
100.5 to	200.5	0.6079697	-1.1029505	1.41567254	1.19337235	3.42826630	100
200.5 to	300.5	-1.5522842	-1.2964018	-0.04143484	-1.03384861	1.07055980	100
300.5 to	400.5	0.2397368	-0.4858291	1.26850188	-0.65888059	2.04322065	100
400.5 to	500.5	-0.3115478	2.0859729	-0.42507561	-0.19530982	0.21883520	100
500.5 to	600.5	0.3223949	-1.2293368	0.59615762	0.60339703	0.71949189	100
600.5 to	700.5	-0.0968183	-0.5421897	0.49023897	-0.46691518	0.45834403	100
700.5 to	800.5	-0.4835011	1.1418296	-1.20257646	2.24331925	6.47867140	100
800.5 to	900.5	1.5691790	-0.6707111	1.01099461	-2.34692427	6.53016362	100
900.5 to	1000.5	0.2647794	0.8368383	-0.14746741	-0.75956943	0.59869235	100
TOTAL Q stats		5.8252263	12.2292054	9.07944973	15.22068253	24.30013226	1000
df for Q stats		5.4442614	8.3570690	7.18372883	9.00000000	16.18372883	0
p-val for Q stats		0.3762832	0.1612299	0.26261685	0.08504985	0.08794875	0

FIGURE: 38

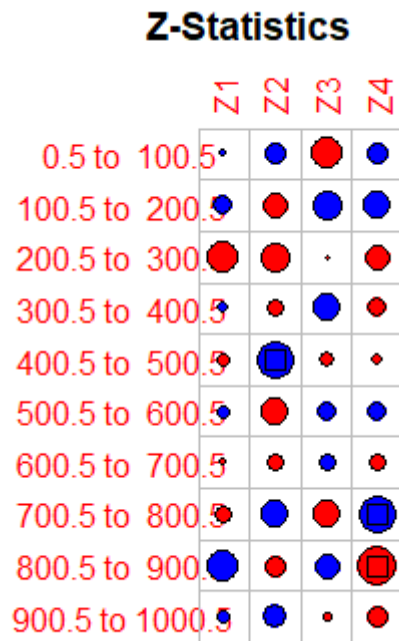


FIGURE: 39

[J]

The BCT model had the best centile curve and that was selected for the handgrip against age dataset for English boys as it covered more data points the BCT Curve also had the lowest GAIC value and was preferred. Examining quantile residuals and Q statistics, the BCT model displays balanced residuals with small deviations from normalcy in a variety of networks. This implies that the BCT model more accurately represents the underlying data distribution. However, there are some noticeable deviations from normalcy from the BCCG and BCPE models within a range, which raises the possibility of limitations in accurately capturing the data's distribution.

Furthermore, a better fit is suggested by the BCT model's worm plot, which has fewer outliers and stays closer to the red line than the other models. This implies that the data is consistently represented by the BCT model and that outliers do not exert an excessive amount of influence. Therefore, the best choice for modelling the data when all analyses—including worm plots,

THIRD DATA SET

[A]

This dataset was obtained from the Kaggle to analyze the relationship between (mpg) which stands for miles per gallon and is used as the response or dependent variable and it was determined from laboratory experiments. This data has 32 obs. of 11 variables. There are no missing variables. The data is in .xlsx format. I have used the Linear regression model to analyze the relationship between the Response variables (mpg) and the Explanatory variables that is Horsepower (hp), Weight(wt.) and Displacement(dis).

[B]

For a variety of data analysis tasks, the MT cars dataset is a flexible resource. We can conduct several analyses with this dataset, including:

- With descriptive statistics, we can determine each variable's mean, median, mode, standard deviation, variance, and range to gain a deeper comprehension of the data.
 - Data visualization: The distribution and relationship between variables can be seen by using a variety of plot types, including scatter plots, histograms, box plots, and bar plots.
 - Correlation analysis: If a relationship exists between two variables, we can ascertain it by computing their correlation coefficient.
- Regression analysis: Using one or more other variables as predictors, we can create regression models to forecast a single variable.

1	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
2	Mazda RX4	21	6	160	110	3.9	2.62	16.46	0	1	4	4
3	Mazda RX4 Wag	21	6	160	110	3.9	2.875	17.02	0	1	4	4
4	Datsun 710	22.8	4	108	93	3.85	2.32	18.61	1	1	4	1
5	Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
6	Hornet Sportabout	18.7	8	360	175	3.15	3.44	17.02	0	0	3	2
7	Valiant	18.1	6	225	105	2.76	3.46	20.22	1	0	3	1
8	Duster 360	14.3	8	360	245	3.21	3.57	15.84	0	0	3	4
9	Merc 240D	24.4	4	146.7	62	3.69	3.19	20	1	0	4	2
10	Merc 230	22.8	4	140.8	95	3.92	3.15	22.9	1	0	4	2
11	Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4
12	Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
13	Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
14	Merc 450SL	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
15	Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18	0	0	3	3
16	Cadillac Fleetwood	10.4	8	472	205	2.93	5.25	17.98	0	0	3	4
17	Lincoln Continental	10.4	8	460	215	3	5.424	17.82	0	0	3	4
18	Chrysler Imperial	14.7	8	440	230	3.23	5.345	17.42	0	0	3	4
19	Fiat 128	32.4	4	78.7	66	4.08	2.2	19.47	1	1	4	1
20	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2

FIGURE: 40

[C]

```
> # Load the mtcars dataset
> data(mtcars)
```

```
> # Display the structure of the dataset
> str(mtcars)
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num   6  6  4  6  8  6  8  4  4  6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num   3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num   0  0  1  1  0  1  0  1  1  1 ...
 $ am  : num   1  1  1  0  0  0  0  0  0  0 ...
 $ gear: num   4  4  4  3  3  3  3  4  4  4 ...
 $ carb: num   4  4  1  1  2  1  4  2  2  4 ...
```

FIGURE: 50

```
> # Display summary statistics
> summary(mtcars)
```

mpg	cyl	disp	hp	drat
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0	Min. :2.760
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5	1st Qu.:3.080
Median :19.20	Median :6.000	Median :196.3	Median :123.0	Median :3.695
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7	Mean :3.597
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0	3rd Qu.:3.920
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0	Max. :4.930

wt	qsec	vs	am	gear
Min. :1.513	Min. :14.50	Min. :0.0000	Min. :0.0000	Min. :3.000
1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:3.000
Median :3.325	Median :17.71	Median :0.0000	Median :0.0000	Median :4.000
Mean :3.217	Mean :17.85	Mean :0.4375	Mean :0.4062	Mean :3.688
3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:4.000
Max. :5.424	Max. :22.90	Max. :1.0000	Max. :1.0000	Max. :5.000

carb
Min. :1.000
1st Qu.:2.000
Median :2.000
Mean :2.812
3rd Qu.:4.000
Max. :8.000

FIGURE: 51

To help comprehend the range, central tendency, and spread of the data, these summary statistics offer a summary of the distribution of each variable in the dataset. The summaries of these variables are comparable, as are their quartile ranges, measures of central tendency(mean and median), and ranges.

```
> # Example scatterplot
> ggplot(mtcars, aes(x = mpg, y = hp)) +
+   geom_point() +
+   labs(x = "Miles Per Gallon", y = "Horsepower")
+ response_variable <- mtcars$mpg
```

FIGURE: 52

From the MT cars dataset, the "Miles Per Gallon" (mpg) variable is represented on the x-axis. From the MT cars dataset, the "Horsepower" (hp) variable is plotted on the y-axis. The weight (wt.) and displacement (disp) variables are additionally displayed in the plot.

A distinct car model is represented by each point on the plot, which illustrates the correlation between those cars' horsepower and miles per gallon.

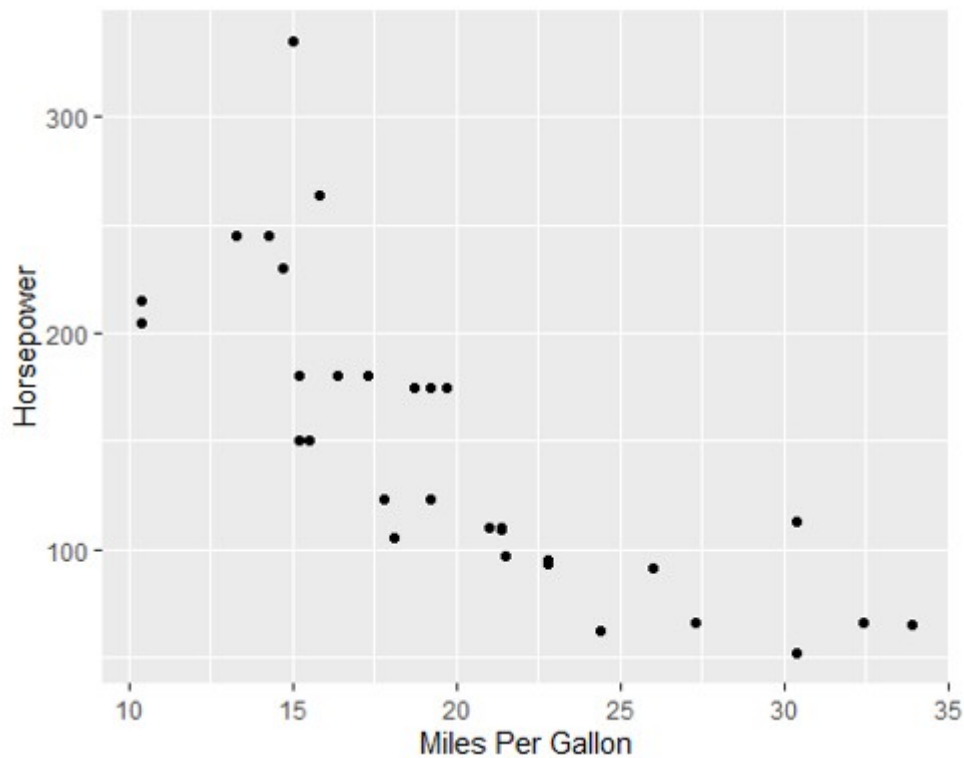


FIGURE: 53

An individualised bin count, fill colour, border colour, transparency, and axis labels are included in this histogram of the "mpg" variable from the MT cars dataset.

```
ggplot(mtcars, aes(x = mpg)) +  
  geom_histogram(bins = 30, fill = "grey", color = "green", alpha = 0.8)+  
  labs(title = "Histogram of mtcars", x = "Miles Per Gallon", y = "Horsepower", "weight", "displacement")
```

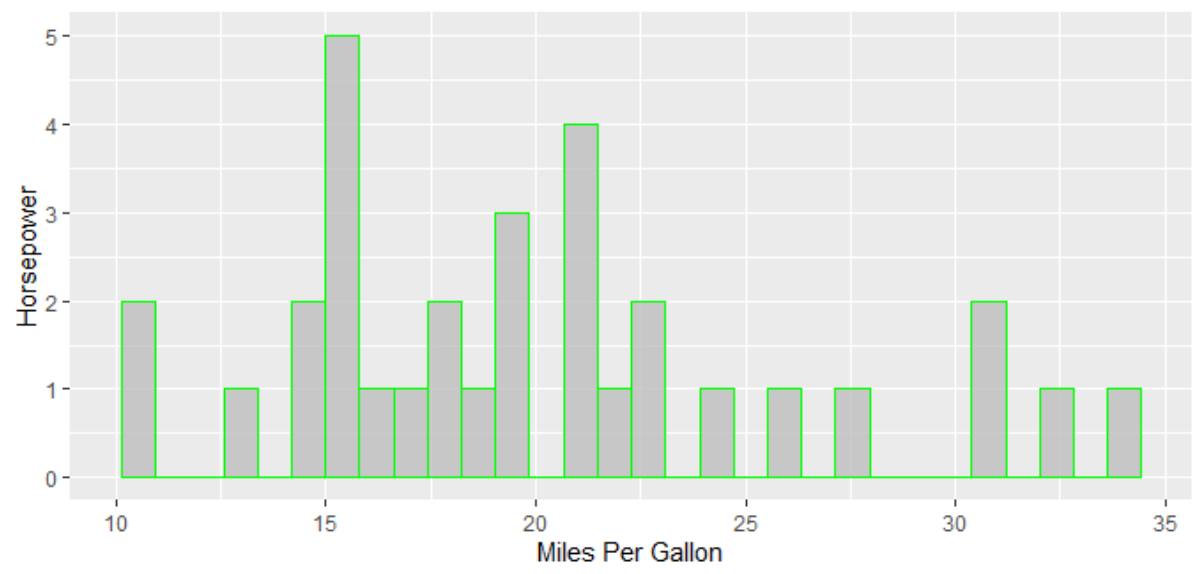


FIGURE:54

The following are the usual themes of this plot Residuals vs Fitted:

The differences between the values that are observed and the values that the model predicts are known as residuals, and they are shown on the vertical axis. The actual data points' deviation from the expected values is shown by the residuals. The fitted values are the predicted values derived from the regression model, and they are represented by the horizontal axis. Based on the independent variables, the dependent variable's estimated values are as follows from the model. About the fitted values, this plot serves to assist you in determining whether the residuals show any systematic behaviour or form a pattern. A model that fits well should have residuals that are dispersed randomly, with no clear pattern, around zero.

```
# Check assumptions
# Example: Residuals vs. Fitted values plot
plot(model, which = 1)
```

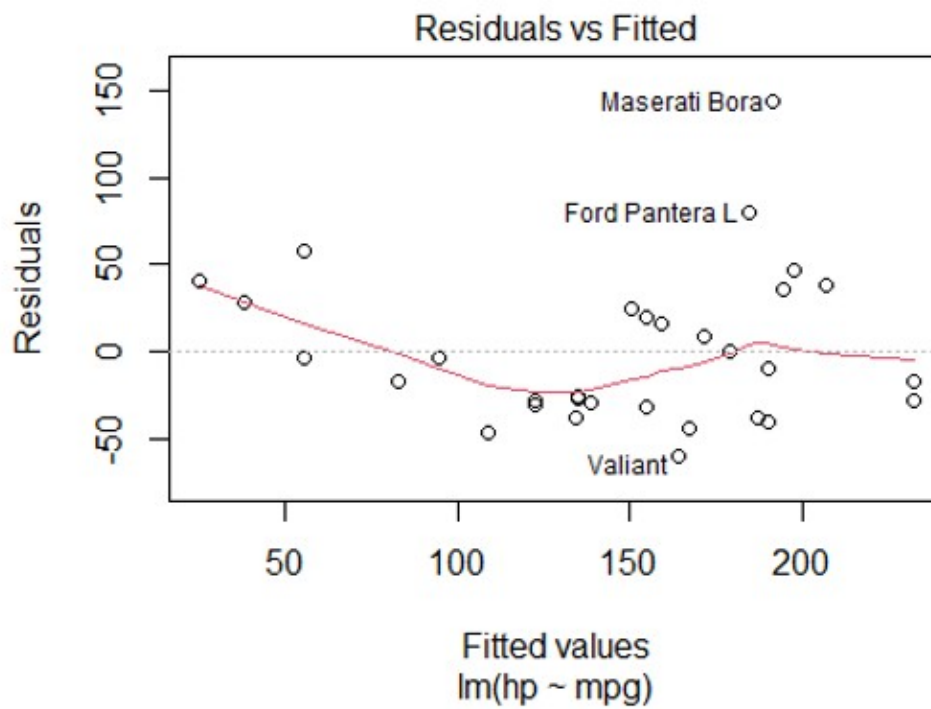


FIGURE: 55

```
ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(bins = 30, fill = "grey", color = "green", alpha = 0.8)+
  labs(title = "Histogram of mtcars", x = "Miles Per Gallon", y = "Horsepower")
```

FIGURE: 56

```
# Define Response and Explanatory variables |
response_variable <- mtcars$mpg
explanatory_variables <- mtcars[,c("Horsepower", "weight", "Displacement")]
```

FIGURE: 57

```

> # Fit a linear regression model
> model <- lm(response_variable ~., data = explanatory_variables)
> # Summary of the model
> summary(model)

Call:
lm(formula = response_variable ~ ., data = explanatory_variables)

Residuals:
    Min       1Q   Median       3Q      Max
-3.891 -1.640 -0.172  1.061  5.861

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.105505   2.110815  17.579 < 2e-16 ***
wt           -3.800891   1.066191  -3.565  0.00133 **
hp           -0.031157   0.011436  -2.724  0.01097 *
disp         -0.000937   0.010350  -0.091  0.92851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.639 on 28 degrees of freedom
Multiple R-squared:  0.8268,    Adjusted R-squared:  0.8083
F-statistic: 44.57 on 3 and 28 DF,  p-value: 8.65e-11

```

FIGURE: 58

The linear regression model seems to be a good fit for the data, based on this R output. According to the R-squared value, the model accounts for a sizable percentage of the response variable's variance, and the p-value of the F-statistic indicates that at least one explanatory variable significantly influences the response variable.

```

> predicted_values <- predict(model, explanatory_variables)
> head(predicted_values)
      Mazda RX4      Mazda RX4 Wag      Datsun 710      Hornet 4 Drive
      23.57003      22.60080      25.28868      21.21667
Hornet Sportabout      Valiant
      18.24072      20.47216

```

FIGURE: 59

```

> library(ggplot2)
> prediction_data <- data.frame(
+   Actual = mtcars$mpg,
+   Predicted = predicted_values
+ )
> str(prediction_data)
'data.frame':   32 obs. of  2 variables:
 $ Actual      : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ Predicted: num  23.6 22.6 25.3 21.2 18.2 ...

```

FIGURE: 60

We may forecast the following based on the given R output:

32 observations, or rows, and 2 variables, or columns, make up the prediction data frame.

The two numerical variables are the Actual and the Predicted.

The actual values of the mpg (miles per gallon) column from the integrated MT cars dataset in R are probably contained in the Actual variable.

The predicted values in the Predicted variable may have come from a model that was trained to forecast mpg values.

CONCLUSION

The first assignment required selecting and fitting the optimal parametric distribution for the data on Dutch boys at 14 age. The location, scale, and shape (Kurtosis and Skewness) of four parameters could be estimated using GAMLSS. After that, I created centile curves using English students' hand grip strength data to evaluate the correlation between hand grip strength and age. Finally, I analysed the data from MT cars using linear regression and model performance evaluation to predict future values.

I was able to assess English schoolboys' Handgrip against age using centile plots for the second coursework assignment. BCCG, BCT, and BCPE were used in this process. The best model was selected after the centile curves for the three models were plotted. After performing the required diagnostics and evaluations, the BCT distribution fit was selected.

I've become more proficient at understanding the datasets and have significantly advanced a number of academic disciplines. Actually, as my extensive research shows, statistical methods are remarkably flexible. I was able to decipher the complexity of BMI change, generate centile curves for hand grip strength, and track MT car data by applying statistical methods to extract actionable data and force an informed decision.

I am learning more about statistical modelling and GAMLSS in general thanks to this coursework. My understanding of the R programming language and how to write code has significantly improved. The information I've gained here will be useful to me later.

REFERENCES

- Lipovetsky, S. (2023). Statistical Modeling of Implicit Functional Relations.
- Robert A. Rigby, M. D. (2020). *Distributions for Modeling Location, Scale, and Shape*. Chapman & Hall.

APPENDIX

```
#Load required libraries
library(gamlss)
```

```
{r}
old <-14
da<-with(dbbmi,subset(dbbmi,age>old&age<old+1))
}
```

```
{r}
data("dbbmi")
}
```

```
{r}
old <-14
da<-with(dbbmi,subset(dbbmi,age>old&age<old+1))
}
```

```
fitGAMLSS_norm <-gamlss(bmi14~1, family = NO())
fitGAMLSS_lognorm <-gamlss(bmi14~1, family = LOGNO())
summary(fitGAMLSS_norm)
# summary
summary(fitGAMLSS_lognorm)

fitnorm <-fitdistr(bmi14, "normal")
fitlognorm <-fitdistr(bmi14, "lognormal")
print(fitnorm)
print(fitlognorm)
```

```
# calculate AIC for each distribution
aic_values <- c(AIC(fit_normal), AIC(fit_lognormal), AIC(fit_gamma), AIC(fit_exponential))
names(aic_values) <- c("Normal", "Log-Normal", "Gamma", "exponential")
# print AIC values
print(aic_values)
```

```
# Fit different distributions
fit_normal <- fitdistr(bmi14, "normal")
fit_lognormal <- fitdistr(bmi14, "lognormal")
fit_exponential <- fitdistr(bmi14, "exponential")
fit_gamma <- fitdistr(bmi14, "gamma")
```

```
ggplot(mydata, aes(x=age, y= grip))+ geom_line(color = "Blue")
```

```
gbccg <-gamlss(grip ~pb(age),sigma.fo = ~pb(age),nu.fo=~pb(age),data = mydata,family = BCCG)
```



```

gbcp <- gamlss(grip ~ pb(age),
              sigma.fo = ~pb(age),
              nu.fo = ~pb(age),
              lambda.fo = ~pb(age),
              data = mydata,
              family = BCPE,
              start.from = gbccg)

edf(gbcp)

gbct <- gamlss(grip ~ pb(age),
              sigma.fo = ~pb(age),
              nu.fo = ~pb(age),
              tau.fo = ~pb(age),
              data = mydata,
              family = BCT,
              start.from = gbccg)

```

```

# Calculate GAIC
gaic_bccg <- GAIC(gbccg)
gaic_bct <- GAIC(gbct)
gaic_bcpe <- GAIC(gbcp)

# Print GAIC values for each model
cat("GAIC for BCCG model:", gaic_bccg, "\n")
cat("GAIC for BCT model:", gaic_bct, "\n")
cat("GAIC for BCPE model:", gaic_bcpe, "\n")

# Compare GAIC values and select the model with the lowest GAIC
if (gaic_bccg < gaic_bct & gaic_bccg < gaic_bcpe) {
  cat("BCCG model has the lowest GAIC and is the preferred model.\n")
} else if (gaic_bct < gaic_bccg & gaic_bct < gaic_bcpe) {
  cat("BCT model has the lowest GAIC and is the preferred model.\n")
} else {
  cat("BCPE model has the lowest GAIC and is the preferred model.\n")
}

```

```

fittedPlot(gbccg, gbct, gbcp, x=mydata$age)
centiles_gbccg <- centiles(gbccg)
centiles_gbct <- centiles(gbct)
centiles_gbcp <- centiles(gbcp)
# Residuals plot
plot(gbccg, main = "BCCG Model Residuals")

```

```

# Fit the models again
library(gamlss)
centiles_split <- list()
for (model in list(gbct, gbcp)) {
  centiles_split[[length(centiles_split) + 1]] <- centiles.split(model, para = "sigma")
}
# worm plot
wp(gbccg)
wp(gbct)
wp(gbcp)

```

```
# Load the mtcars dataset
data(mtcars)

# Display the structure of the dataset
str(mtcars)

# Display summary statistics
summary(mtcars)

# Example scatterplot
ggplot(mtcars, aes(x = mpg, y = hp , wt, disp)) +
  geom_point() +
  labs(x = "Miles Per Gallon", y = "Horsepower", "weight", "Displacement")
# Define Response and Explanatory Variables
response_variable <- mtcars$mpg
explanatory_variables <- mtcars[,c("Horsepower", "weight", "Displacement")]

# Fit a linear regression model
model <- lm(response_variable ~. , data = explanatory_variables)
summary(model)

# Summary of the model
summary(model)

# Check assumptions
# Example: Residuals vs. Fitted values plot
plot(model, which = 1)
ss
# Example: Normal Q-Q plot
plot(model, which = 2)
```